

MAT1002 Lecture 8,

Outline

- Sketching polar curves
- Polar curves : arc lengths and areas
- The 3D space
- Vectors

Polar Curves

Symmetry

Suppose we want to sketch a polar curve $r = f(\theta)$ (or more generally $F(r, \theta) = 0$) on the xy -plane. Identifying symmetry may save some effort.

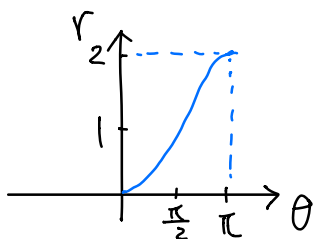
Symmetry about x -axis:
 (r, θ) on $C \Rightarrow (r, -\theta)$ or $(-r, -\theta + \pi)$ on C

Symmetry about y -axis:
 (r, θ) on $C \Rightarrow (r, \pi - \theta)$ or $(-r, -\theta)$ on C

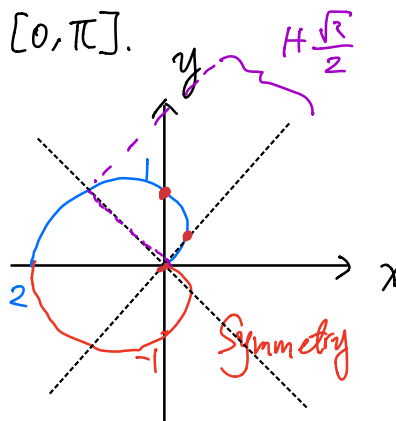
Symmetry about x -axis:
 (r, θ) on $C \Rightarrow (r, \theta + \pi)$ or $(-r, \theta)$ on C

e.g. Sketch $r = 1 - \cos \theta$ (Cardioid) on the xy -plane.

Sol: Since $r = 1 - \cos \theta = 1 - \cos(-\theta)$, curve is symmetric about y -axis: only need to investigate $\theta \in [0, \pi]$.

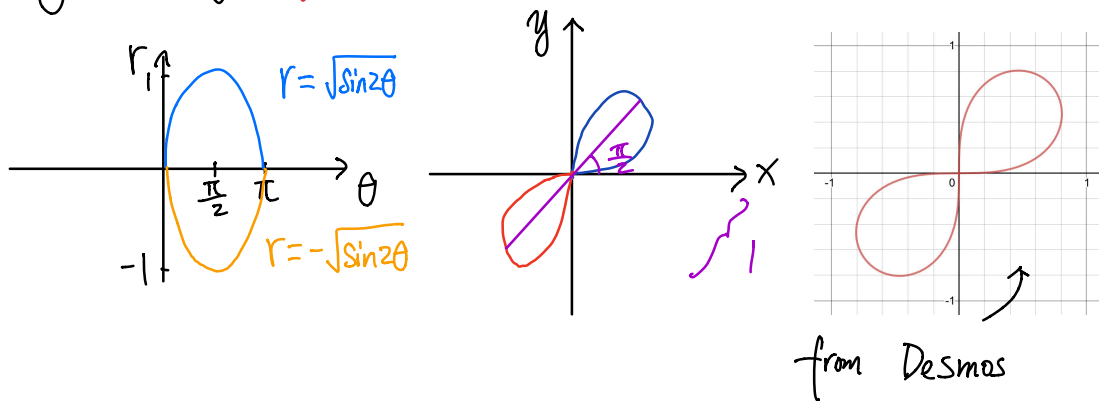


θ	r
0	0
$\frac{\pi}{4}$	$1 - \frac{\sqrt{2}}{2}$
$\frac{\pi}{2}$	1
$\frac{3\pi}{4}$	$1 + \frac{\sqrt{2}}{2}$
π	2



e.g. How does the polar curve $r^2 = \sin 2\theta$ look like on the (x, y) -plane?

- Since $(r)^2 = r^2$, curve is symmetric about the origin.
- For $\theta \in [0, 2\pi]$, $\sin 2\theta \geq 0 \Leftrightarrow \theta \in [0, \frac{\pi}{2}] \cup [\pi, \frac{3\pi}{2}]$.
- By symmetry, suffices to investigate $\theta \in [0, \frac{\pi}{2}]$.



Slope

For a polar curve $r = f(\theta)$, the xy -positions of its points are

$$x = f(\theta) \cos \theta, \quad y = f(\theta) \sin \theta,$$

with angle θ being the parameter. The slope at a point given by $\theta = \theta_0$ is

$$\left. \frac{dy}{dx} \right|_{\theta = \theta_0} = \frac{f(\theta) \cos \theta + f'(\theta) \sin \theta}{-f(\theta) \sin \theta + f'(\theta) \cos \theta} \Big|_{\theta = \theta_0},$$

given that the denominator $\neq 0$.

Arc Lengths

Consider a curve on the xy -plane given in polar coordinates by

$$\underline{r = f(\theta)}, \quad a \leq \theta \leq b,$$

where f' is continuous. If the curve is traversed exactly once, then by the arc length formula ~~on Page 8~~, its length L is given by

$$L = \int_a^b \sqrt{(f(\theta))^2 + f'(\theta)^2} d\theta = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta.$$

To see this, since $x = f(\theta)\cos\theta$, $y = f(\theta)\sin\theta$,

$$\begin{aligned} L &= \int_a^b \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta \\ &= \int_a^b \left((f'(\theta)\cos\theta - f(\theta)\sin\theta)^2 + (f'(\theta)\sin\theta + f(\theta)\cos\theta)^2 \right)^{\frac{1}{2}} d\theta \\ &= \int_a^b \left(f'(\theta)^2\cos^2\theta + f(\theta)^2\sin^2\theta + f'(\theta)^2\sin^2\theta + f(\theta)^2\cos^2\theta \right)^{\frac{1}{2}} d\theta \\ &= \int_a^b \sqrt{f(\theta)^2 + f'(\theta)^2} d\theta \end{aligned}$$

e.g. Find the arc length of the polar curve $r = 1 - \cos\theta$.

Sol. Curve is traced out exactly once for $0 \leq \theta \leq 2\pi$.

$$\begin{aligned} L &= \int_0^{2\pi} \sqrt{(1 - \cos\theta)^2 + \sin^2\theta} d\theta = \int_0^{2\pi} \sqrt{2 - 2\cos\theta} d\theta \\ &= \sqrt{2} \int_0^{2\pi} \sqrt{1 - \cos\theta} d\theta = \sqrt{2} \int_0^{2\pi} \sqrt{2\sin^2\frac{\theta}{2}} d\theta \quad \left(\begin{array}{l} \cos 2x \\ = 1 - 2\sin^2 x \end{array} \right) \end{aligned}$$

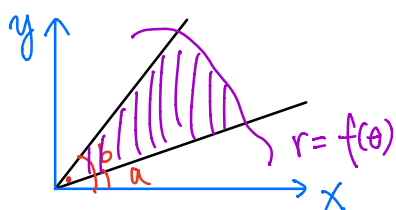
$$= \underline{\underline{\sqrt{2}\sqrt{2}}} \int_0^{2\pi} \underline{\underline{\sin \frac{\theta}{2}}} d\theta = 2 \left(2 \cos \frac{\theta}{2} \right) \Big|_{2\pi}^0 = 4(1) = 8.$$

Areas: Fan-Shaped Regions

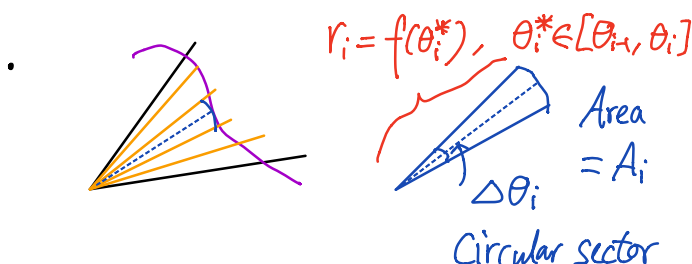
Consider the region on the xy -plane whose polar coordinates satisfy

$$0 \leq r \leq f(\theta), \quad a \leq \theta \leq b \quad (b-a \leq 2\pi).$$

This is a fan-shaped region. What is its area A ?



• Partition the θ -interval $[a, b]$.

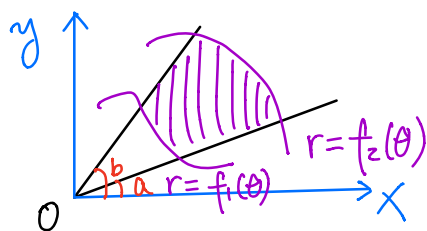


$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n A_i = \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{\Delta\theta_i}{2\pi} \pi f(\theta_i^*)^2 = \lim_{n \rightarrow \infty} \frac{1}{2} \sum_{i=1}^n f(\theta_i^*)^2 \Delta\theta_i$$

$$A = \frac{1}{2} \int_a^b f(\theta)^2 d\theta.$$

More generally, for the xy -plane region whose polar coordinates satisfy

$$0 \leq f_1(\theta) \leq r \leq f_2(\theta), \quad a \leq \theta \leq b \quad (b-a \leq 2\pi),$$



its area is given by

$$A = \frac{1}{2} \int_a^b (f_2(\theta)^2 - f_1(\theta)^2) d\theta.$$

Example

- (a) Find the area of the region enclosed by the curve

$$r = 2(1 + \cos \theta), \quad 0 \leq \theta \leq 2\pi.$$

- (b) Consider the region S enclosed by the curve

$$r = 1 - \cos \theta, \quad 0 \leq \theta \leq 2\pi.$$

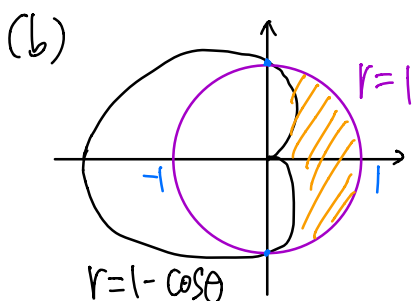
Find the area of the region that lies outside S and inside the circle $r = 1$.

Solution

- (a) 6π .

- (b) $2 - (\pi/4)$.

Exercise.



$$\begin{aligned} \cos 2x &= \cos^2 x - \sin^2 x \\ &= 2\cos^2 x - 1 \end{aligned}$$

$$1 - \cos \theta \leq r \leq 1, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$

$$A = \int_{-\pi/2}^{\pi/2} \frac{1}{2} (1 - (1 - \cos \theta)^2) d\theta$$

$$= \frac{1}{2} \int_{-\pi/2}^{\pi/2} (2 \cos \theta - \cos^2 \theta) d\theta$$

$$= \int_0^{\pi/2} (2 \cos \theta - \cos^2 \theta) d\theta$$

$$= \int_0^{\pi/2} \left(2 \cos \theta - \frac{1 + \cos 2\theta}{2} \right) d\theta$$

$$= \dots = 2 - \frac{\pi}{4}.$$

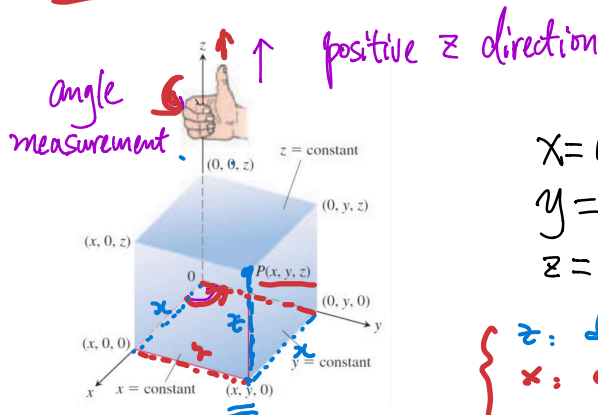
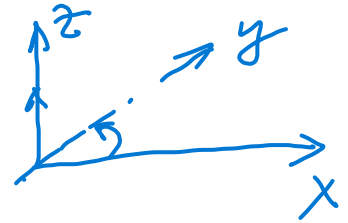
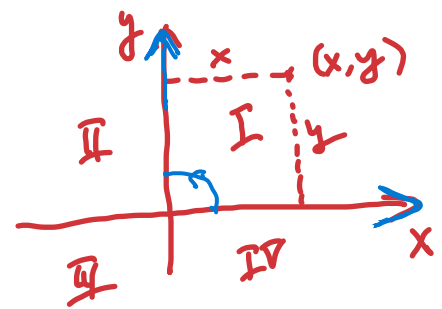
The 3D Space

Coordinate Axes and Their Placement

We will represent the (three-dimensional) space using $\mathbb{R}^3$, where

$$\mathbb{R}^3 := \{(x, y, z) : x \in \mathbb{R}, y \in \mathbb{R}, z \in \mathbb{R}\}.$$

The number x , y and z represent the position of the point $P(x, y, z)$, with respect to the coordinate axes (x-axis, y-axis and z-axis). The coordinate axes are usually arranged according to the right-hand rule, as shown in the following figure.



$x=0$: yz -plane

$y=0$: xz -plane

$z=0$: xy -plane

Coordinate planes.

z : distance to xy -plane
 x : distance to yz plane
 y : distance to xz plane.

First octant :

$$\{(x, y, z) : x > 0, y > 0, z > 0\}.$$

The set of all points (x, y, z) which satisfy

$$x^2 + y^2 = 4, z = 3$$

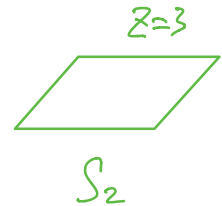
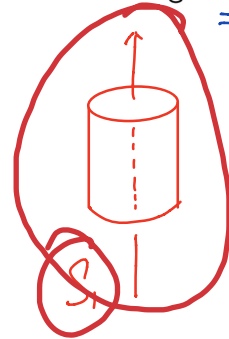
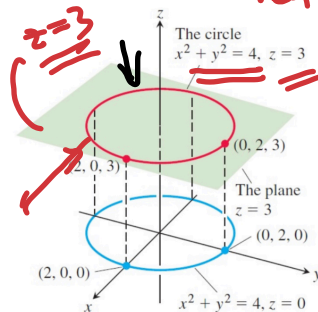
$$\{(x, y, z) : x^2 + y^2 = 4 \text{ \& } z = 3\} \\ = \{(x, y, z) : x^2 + y^2 = 4\} \cap \{(x, y, z) : z = 3\}$$

S_1 : cylinder

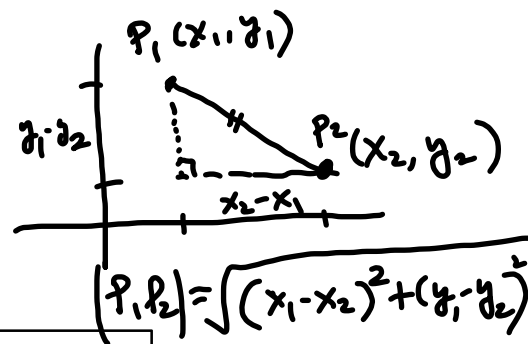
can be represented geometrically by the pink disk in the following figure.

$$= S_1 \cap S_2$$

S_2 : plane



Distance & Spheres



Definition

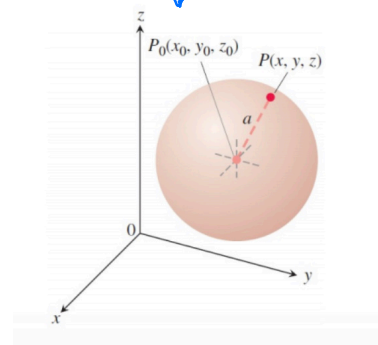
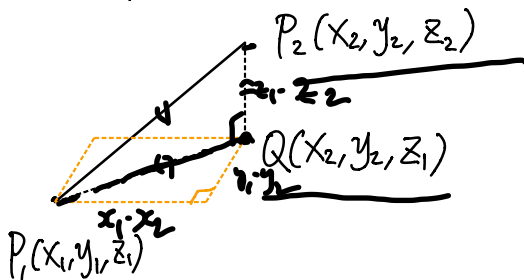
The distance $|P_1 P_2|$ between the points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is defined by

$$|P_1 P_2| := \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

The sphere with radius a centered at the point (x_0, y_0, z_0) is the set of all points (x, y, z) that satisfy

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = a^2.$$

The definition of $|P_1 P_2|$ can be understood as follows.



$$|P_1 P_2| = \sqrt{|P_1 Q|^2 + |P_2 Q|^2} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Generalizing from 1D, 2D and 3D cases, the distance between

$P_1 := (x_1, x_2, \dots, x_n)$ and $P_2 := (y_1, y_2, \dots, y_n)$ in $\mathbb{R}^n$

is

$$|P_1 P_2| := \sqrt{(y_1 - x_1)^2 + \dots + (y_n - x_n)^2} = \left(\sum_{i=1}^n (y_i - x_i)^2 \right)^{\frac{1}{2}}.$$

Example

Show that $x^2 + y^2 + z^2 + 4x - 6y + 2z + 6 = 0$ is the equation of a sphere, and find its center and radius.

$$\begin{aligned} \text{Sol. } (*) &\Leftrightarrow (x^2 + 4x + 4) + (y^2 - 6y + 9) + (z^2 + 2z + 1) - 14 + 6 = 0 \\ &\Leftrightarrow (x+2)^2 + (y-3)^2 + (z+1)^2 = (\sqrt{8})^2. \end{aligned}$$

Center: $(-2, 3, -1)$; radius: $2\sqrt{2}$.

Remark With center (x_0, y_0, z_0) and radius a :

$$(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2 = a^2 : \text{Sphere.}$$

$$(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2 \leq a^2 : \text{Closed ball. (Interior + sphere.)}$$

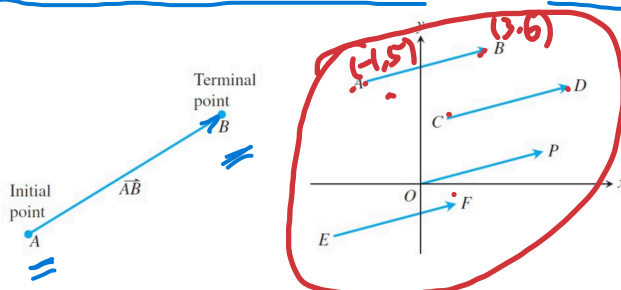
$$(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2 < a^2 : \text{Open ball. (Interior.)}$$

Vectors

speed

Roughly speaking, a **vector** is a directed line segment that goes from a point A (the **initial point**) to a point B (the **terminal point**). Such a vector is denoted by $\overrightarrow{AB}$.

- ▶ The two points A and B can be the same. $\overrightarrow{AA}$: zero vector.
- ▶ There are two defining properties of a vector: **length** and **direction**.
- ▶ Two vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are considered exactly the same if they have the same length and direction, even if $A \neq C$.



$\overrightarrow{AB}$ is also written as

$\overrightarrow{AB}$.

$$\begin{aligned} O &= (0, 0) \\ P &= (4, 1) \end{aligned}$$

$$\overrightarrow{OP} = (4, 1)$$

$$A = (-1.5)$$

$$B = (3, 6)$$

$$\overrightarrow{AB} = (4, 1)$$

$$\underline{\underline{\vec{CD}}} = (\underline{x_3 - x_4}, \underline{y_3 - y_4})$$

All of the same vectors above can be represented by $\vec{OP}$, the one that starts at the origin O .

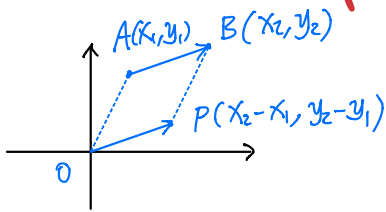
$$\underline{\underline{\vec{AB}}} = (\underline{x_2 - x_1}, \underline{y_2 - y_1})$$

Let $\vec{AB}$ be a vector in the xy -plane, where $A = (x_1, y_1)$ and $B = (x_2, y_2)$. Consider translating A to the origin O . It is not hard to see that $\vec{OP} = \vec{AB}$ if and only if

$$P = (x_2 - x_1, y_2 - y_1).$$

Therefore $\vec{AB}$ can be recorded by the position of P alone, and we write

$$\underline{\underline{\vec{AB}}} = \langle \underline{x_2 - x_1}, \underline{y_2 - y_1} \rangle \text{ or } \vec{AB} = \begin{bmatrix} x_2 - x_1 \\ y_2 - y_1 \end{bmatrix}.$$



If $A = (x_1, y_1, z_1)$ and $B = (x_2, y_2, z_2)$ are points in the 3D space, then $\vec{AB} = \vec{OP}$, where $P = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$.

In this case, we write

$$\boxed{\vec{AB}} = \langle \underline{x_2 - x_1}, \underline{y_2 - y_1}, \underline{z_2 - z_1} \rangle \text{ or } \vec{AB} = \begin{bmatrix} x_2 - x_1 \\ y_2 - y_1 \\ z_2 - z_1 \end{bmatrix}.$$

$\quad \quad \quad v_1 \quad v_2 \quad v_3$

$$D(A, B) = \sqrt{v_1^2 + v_2^2 + v_3^2}$$

$\vec{v}$

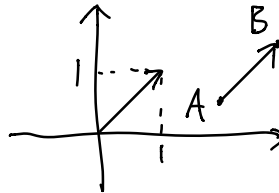
Component Forms and Length

- ▶ In general, every vector in the xy -plane can be represented as a vector $\vec{v} = \langle v_1, v_2 \rangle$. The numbers v_1 and v_2 are called the components of $\vec{v}$.
or $\vec{v}$ ← Usually used for printing.
 - ▶ A common alternative notation for $\vec{v}$ is $\mathbf{v}$.
 - ▶ The length of the vector $\vec{v} = \langle v_1, v_2 \rangle$ is defined to be
(or norm or magnitude) $\sqrt{v_1^2 + v_2^2}$.
- The length of $\vec{v}$ is commonly denoted by $|\vec{v}|$ or $\|\vec{v}\|$.
- ▶ Similarly, every vector in the space can be written as $\vec{v} = \langle v_1, v_2, v_3 \rangle$, with

$$|\vec{v}| := \|\vec{v}\| := \sqrt{v_1^2 + v_2^2 + v_3^2}.$$

More generally, we can talk about d -dimensional vectors, and each such vector $\vec{v}$ can be represented by $\langle v_1, v_2, \dots, v_n \rangle$,
where $v_1, \dots, v_n$ are the components of $\vec{v}$.

where $|\vec{v}| := \|\vec{v}\| := \sqrt{v_1^2 + \dots + v_n^2}$, although they are hard to draw for $d \geq 4$.



$$\vec{AB} = \langle 1, 1 \rangle.$$

Algebraic Operations of Vectors

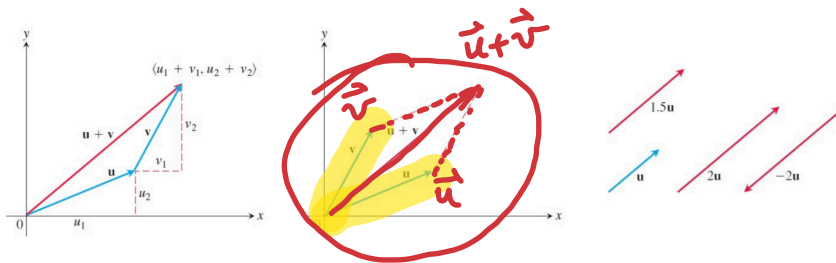
Definition

Given two vectors $\vec{u} = \langle u_1, u_2, \dots, u_n \rangle$, $\vec{v} = \langle v_1, v_2, \dots, v_n \rangle$ and a scalar $k \in \mathbb{R}$, define additions and scalar multiplications as follows:

- ▶ $\vec{u} + \vec{v} := \langle u_1 + v_1, u_2 + v_2, \dots, u_n + v_n \rangle$.
- ▶ $k\vec{u} := \langle ku_1, ku_2, \dots, ku_n \rangle$.

Similar to real numbers, we write $\vec{u} - \vec{v}$ to mean $\vec{u} + (-1)\vec{v}$.

^{geometric}
The ~~geometry~~ effect of these operations on vectors in $\mathbb{R}^2$ is displayed in the following figure.



• $(-1)\vec{v}$ is pointing to the "opposite" direction as $\vec{v}$.

• For $\vec{v}$ in $\mathbb{R}^n$,

$$|k\vec{v}| = \sqrt{(kv_1)^2 + \dots + (kv_n)^2} = |k| \sqrt{v_1^2 + \dots + v_n^2} = |k| |\vec{v}|,$$

So length of $k\vec{v}$ is $|k|$ times that of $\vec{v}$.

Properties of Algebraic Operations

Definition

The zero vector, denoted by $\vec{0}$ or $\mathbf{0}$, is the vector whose components are all zeros.

Properties of Vector Operations //

Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be vectors and a, b be scalars.

Same dimension

1. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$

3. $\mathbf{u} + \mathbf{0} = \mathbf{u}$

5. $0\mathbf{u} = \mathbf{0}$

7. $a(b\mathbf{u}) = (ab)\mathbf{u}$

9. $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$

2. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$

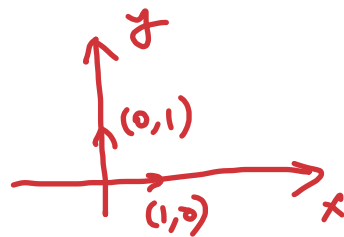
4. $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$

6. $1\mathbf{u} = \mathbf{u}$

8. $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$

Proof is straight-forward (direct verification).

Unit Vectors and Standard Unit Vectors



Definition

- ▶ A **unit vector** is a vector whose length is 1.
- ▶ The **standard unit vectors in $\mathbb{R}^3$** are the unit vectors

$$\vec{i} := \mathbf{i} := \langle 1, 0, 0 \rangle, \quad \vec{j} := \mathbf{j} := \langle 0, 1, 0 \rangle, \quad \vec{k} := \mathbf{k} := \langle 0, 0, 1 \rangle.$$

Remark

- ▶ Any vector $\vec{v} = \langle v_1, v_2, v_3 \rangle$ in $\mathbb{R}^3$ can be written as a linear combination of $\vec{i}$, $\vec{j}$ and $\vec{k}$:

$$\vec{v} = v_1 \vec{i} + v_2 \vec{j} + v_3 \vec{k}.$$

- ▶ Whenever $|\vec{v}| \neq 0$, the vector $\vec{v}/|\vec{v}|$ is a unit vector, which is called the **direction** of $\vec{v}$.

$$\langle v_1, v_2, v_3 \rangle = \langle v_1, 0, 0 \rangle + \langle 0, v_2, 0 \rangle + \langle 0, 0, v_3 \rangle$$

$$= v_1 \vec{i} + v_2 \vec{j} + v_3 \vec{k}$$

$$+ v_3 \vec{k}$$

$$\vec{v} = \langle 1, 3.5 \rangle$$

$$|\vec{v}| = \sqrt{1 + 3^2 + 5^2} = \sqrt{35}$$

$$\vec{u} = \frac{\vec{v}}{|\vec{v}|} = \frac{1}{\sqrt{35}} \langle 1, 3.5 \rangle$$

$$|\vec{u}| = 1$$

$$\left| \frac{1}{|\vec{v}|} \vec{v} \right| = \frac{1}{|\vec{v}|} |\vec{v}| = 1$$

For any nonzero vector $\vec{v}$, $\vec{v} = \underbrace{|\vec{v}|}_{\text{length}} \underbrace{\left(\frac{\vec{v}}{|\vec{v}|} \right)}_{\text{direction}}$

e.g. If $\vec{v} = \langle 3, -4 \rangle$ is the velocity vector, then

$|\vec{v}| = 5$ is the **speed**, and $\langle \frac{3}{5}, \frac{-4}{5} \rangle$ is the **direction** of the motion.

$$\vec{u} = \langle \frac{3}{5}, \frac{-4}{5} \rangle \text{ unit vector}$$

$$\vec{u} = \frac{\vec{v}}{|\vec{v}|}$$

unit vector